

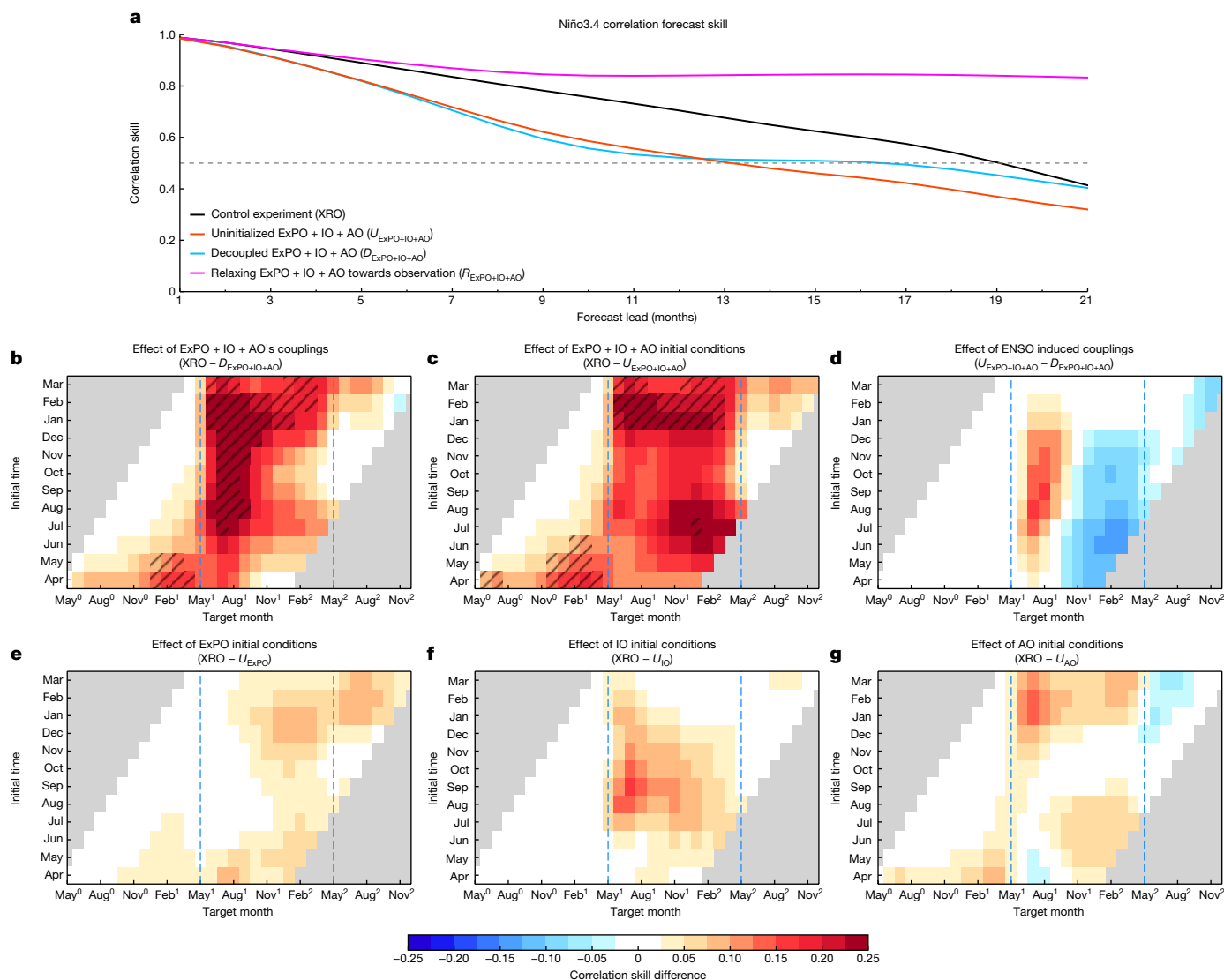


Fig. 3 | Quantifying the increased ENSO forecast skills from the coupled influences outside equatorial Pacific during 1979–2022. a, The all-months correlation skill of the 3 month running mean Niño3.4 index as a function of the forecast lead month in the control experiment (XRO, black line), the uninitialized ExPO + IO + AO experiment ($U_{\text{ExPO}+\text{IO}+\text{AO}}$, removing initial conditions of other basins; red line), the decoupled ExPO + IO + AO experiment ($D_{\text{ExPO}+\text{IO}+\text{AO}}$, removing the coupling of ENSO with other basins; blue line) and the relaxing ExPO + IO + AO to observations experiment ($R_{\text{ExPO}+\text{IO}+\text{AO}}$, adding perfect ‘future’ information of other basins in a hindcast case; magenta line). **b–d**, The skill difference of the Niño3.4 index as a function of initial time and target

month between XRO and $D_{\text{ExPO}+\text{IO}+\text{AO}}$ (**b**), between XRO and $U_{\text{ExPO}+\text{IO}+\text{AO}}$ (**c**) and between $U_{\text{ExPO}+\text{IO}+\text{AO}}$ and $D_{\text{ExPO}+\text{IO}+\text{AO}}$ (**d**). **e–g**, Same as **d**, but for the difference between control and the uninitialized ExPO (**e**), IO (**f**) and AO (**g**) experiments. Hatching indicates that the correlation difference is significant at a 90% confidence level using the two-tailed Fisher z-transformation test. The sensitivity experiments demonstrate the importance of the extratropical Pacific, IO and AO in enhancing ENSO forecast skill, with distinct seasonal dependence. The interbasin memory sustains ENSO forecast skill beyond the SPB with the IO and AO contributing skill in boreal summer and the ExPO in boreal winter.

followed by sequential WWV discharge, which aligns with the finding that ENSO is thermally driven by the SPM³⁷ (Extended Data Fig. 6a,b).

We also find that coupling with the NPM tends to favour multi-year ENSO events, such as the 1998–2000 La Niña. The first year La Niña in 1998–1999 set the stage for a strong spring NPM cooling in 1999 (consistent with the strong nearly instantaneous feedback mechanism³⁸), which in turn reinforced WWV discharge and colder SSTAs (by roughly 0.3 °C) in the second year. This strong WWV discharged state persisted and re-intensified into the third year, causing SSTA to decrease (roughly 0.4 °C) in the winter of the third year (Fig. 4d). Similar patterns are evident in multi-year La Niña events in 2007–2008, 2010–2011 and 2020–2022 (blue shading in Extended Data Fig. 6a). We emphasize that this contribution is also evident for the opposite ENSO phase, as seen in multi-year El Niño events in 1986–1987, 2014–2015 and 2018–2019 (Extended Data Fig. 6a). These results support the hypothesis that the

coupling between NPM and ENSO favours the existence of multi-year ENSO events^{39–41}.

In the IO, the 1996 boreal autumn negative IOD event was found to induce a roughly 0.4 °C Niño3.4 SSTA increase roughly 15 months later, thus contributing to the 1997–1998 super El Niño (Fig. 4f). Conversely, the 1997 boreal autumn positive IOD event led to a roughly 0.5 °C Niño3.4 SSTA decrease roughly 15 months later, thus playing a role in the 1998–1999 La Niña (Fig. 4f). This aligns with a previous finding¹⁵ that a negative IOD event favours the build-up of WWV (that is, recharge) and contributes to the development of El Niño in the following year by means of the Bjerknes feedback. The SIOD mode, characterized by an SST east–west dipole over the southern IO, tends to induce a roughly 0.2 °C Niño3.4 SSTA increase or decrease roughly 12–16 months later, often offsetting the IOD’s effect (Fig. 4g). The IOB, although largely forced by ENSO, helps to accelerate the phase transition of